

INTERNAL GOLDEN OPERATIONAL REFERENCE · DOCUMENTARY RECONSTRUCTION

ELISENCE PHASE 4

GOLDEN CAPABILITY & ARCHITECTURE REFERENCE

Authoritative internal documentary reconstruction for operational reactivation and evidence use

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Golden Capability Ledger rows: 62

Note: Status wording in this document is documentary only. This PDF is a rendered companion to the Markdown authority.

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- 7. ANALYTICS & HEALTH INTELLIGENCE
 - 8. POPULATION HEALTH & EFFECTIVENESS
 - 9. MEDICATION / TERMINOLOGY / INTEROPERABILITY
 - 10. RESEARCH & EVIDENCE
 - 11. PUBLIC / GOVERNMENT / INSTITUTIONAL CAPABILITY
 - 12. PRIVACY ARCHITECTURE
 - 13. ZERO TRUST / RBAC / AUTHORITY
 - 14. AUDIT / PROVENANCE / WORM
 - 15. DATA CONTRACTS / SCHEMA / VERSIONING
 - 16. CONNECTORS / GLOBAL INSIGHTS
 - 17. RESILIENCE / FAILURE SAFETY
 - 18. OBSERVABILITY / MONITORING
 - 19. SYNTHETIC DATA / TEST INFRASTRUCTURE
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GOLDEN CAPABILITY & ARCHITECTURE REFERENCE

Document type: Authoritative internal documentary reconstruction

Mission: ELISENCE_PHASE4_GOLDEN_CAPABILITY_AND_ARCHITECTURE_RECONSTRUCTION_V1

Source root: /Users/elisenceclean/Downloads/ELISENCE_PHASE4_PHASE8/PHASE4

Supporting prior report:

OUTPUTS/PHASE4_REVIEW/ELISENCE_PHASE4_COMPLETE_CAPABILITY_ARCHITECTURE_TEST_EVIDENCE_AND_INTEGRATION_READINESS_REPORT.md

This file: golden checklist for strategic evidence **and** future reactivation

Code review: not performed (documentary mission)

Source files modified: 0

Status wording used below is documentary only. Absence of code in this folder is **not** treated as proof of non-implementation. Where code was not inspected, reactivation fields remain future checkboxes.

1. EXECUTIVE SYSTEM DEFINITION

ELISENCE Phase 4, as reconstructed from the historical documentary corpus, is **not a single thin feature**. It is a **system layer** for turning operational health/medication/lifestyle signals into **privacy-preserving, auditable, multilingual, institution-grade analytics and exchange capability**.

The corpus contains **two major engineering identities** plus one **investor intelligence narrative**:

Identity	Documentary name	System character
A	Global Analytics & Evidence Engine	12-module analytics/privacy/RBAC/WORM/resilience/AC harness stack; claims architecture+implementation completed and ~244 tests passing (Phase4-Report.pdf)
B	GHX — Global Health Exchange & Intelligence Gateway	Jurisdiction/consent/FHIR/CDS/warehouse/FL/digital-twin/ministry-hospital-research product architecture with BBVP M1–M10 and DONE=100%+Evidence (Phase4-GHX.docx, سندGHX.pdf)
C	Stability / Integrity / Comparative Intelligence Layer (often ALISEN-branded)	Trust pillars + Global Medical Intelligence Engine (effectiveness/safety/comparative/NBI) (Phase4-investor report.pdf, Phas4-investor issue report.pdf)

Golden rule for future teams: treat A/B/C as related but not automatically identical packages until code reconciliation proves otherwise. Do **not** drop capabilities from any identity during reactivation.

Cross-cutting non-negotiables recurring across identities: - Non-diagnostic / non-device posture for citizen-facing analytics/CDS - Privacy-preserving aggregates (k-anonymity / DP / AKAC) - Auditability (WORM / provenance) - Multilingual EN/FA/AR/TR/RO with RTL for FA/AR - Zero-Trust / role-bound access - Resilience / fail-safe behaviour

2. ORIGINAL PHASE 4 MISSION

From Phase4-Report.pdf §1:

It exists because operational Phase 3 care/medication activity alone does not produce **policy-grade, research-grade, privacy-safe evidence**. Phase 4 sits *on top of* operational systems, **read-only** to Phase 3, building aggregates/trends/KPIs **without writing back**.

GHX documents expand the mission to a **national-grade exchange/governance backbone** so Elisence can interoperate with hospitals/ministries/research under jurisdiction-aware consent and standards (سندGHX.pdf, Phase4-GHX.docx).

Investor documents frame Phase 4 as the **trust engine** without which later phases cannot be nationally credible (Phas4-investor issue report.pdf).

3. HISTORICAL PROGRAMME CONTEXT

- Built under **Zero-Trust, BBVP**, regulatory-alignment language (NHS/GDPR/MDR Rule 11 non-diagnostic/FDA-EMA RWE/GCC PDPL).
- Multi-channel inputs: sensor, voice, manual (Phase4-Report.pdf).
- Relationship to earlier phases: Phase 3 = transactional care/medication; Phase 4 = evidence/intelligence layer.
- GHX lists upstream/governed engines including phases 3,7,8,9,10,12,15,16,17 (سندGHX.pdf).
- Post-Phase 8 roadmap later defines a freeze→scan→rebuild merge procedure (Plan after cores and engines. Phase 8.pdf) — useful for reactivation order, **not** proof of present mounting.
- Misfile warning: Phase4 Excutive.pdf is **Phase 6** medication-brain content; excluded from Phase 4 capability ledger except as contamination note.

4. COMPLETE SYSTEM ARCHITECTURE

4.1 Analytics Engine data flow (Identity A)

- Ingest from Phase 3 / external (read-only)
- Vocabulary mapping
- Aggregation KPI compute (utilization/effectiveness/satisfaction/adherence/AE)
- Privacy Guard + RBAC
- Consume via Global Insights APIs / BI / PDF export
- Monitor metrics + WORM evidence; Go/No-Go before deployment

4.2 GHX 4+1 tier architecture (Identity B)

- Tier 0:** Identity, Jurisdiction, Locale/Culture, Policy Registry
- Tier 1:** ISecureDataEmitter → SecureWrapper → FHIRTransformer → FHIRValidator → Ledger → GHXQueue
- Tier 2:** Consent, AKAC/DP, RBAC/ABAC, CDS Hooks
- Tier 3:** Jurisdiction warehouses, federated query, federated learning
- Tier 4:** Ministry portal, hospital/clinic integration, research/OMOP feeds, digital twin APIs

4.3 Trust boundaries

- Phase 3 write isolation (Analytics)
- Consent/jurisdiction/mode gates before export (GHX)
- Public vs Research vs Admin (Analytics) and Citizen/Clinician/Ministry/Research modes (GHX)
- Feature-flag hard OFF (GHX BBVP)

5. SYSTEM / CORE / ENGINE MASTER MAP

Identity A — 12 modules

- Data Vocabulary & Mapping Layer

2. Utilization Aggregation Engine

3. Effectiveness Engine (Δ Weight 30/60/90)
- FUSENCE Phase 4 – Golden Capability & Architecture Reference

Internal Golden Operational Reference
4. Satisfaction / Adherence / AE Radar Aggregation (three engines)

5. Privacy Guard (k-Anonymity + DP)

6. RBAC (Public/Research/Admin)

7. Global Insights UI Connectors

8. Monitoring, Metrics & WORM Ledger

9. Quality Gates (Data Contracts, JSON Schemas, Synthetic Data)

10. Failure Modes & Resilience (Circuit Breaker, SLG, Read-Only, Backfill)

11. AC-1..AC-20 Acceptance Criteria Engine + Go/No-Go Harness

12. Multi-Language PDF Export Engine

Identity B — 20 named GHX components (Deck)

GHX Foundation; Locale & Culture; Global Policy Registry; ISecureDataEmitter; SecureWrapper; FHIRTransformer; FHIRValidator; ConsentEngine; Privacy/AKAC+DP; Actors & Modes; CulturalBaseline & i18n; WORM/Ledger; GHXQueue; Warehouse & Federated Query; Digital Twin Orchestration; CDS Hooks; Government Portal; Hospital Integration; Research & OMOP Export; Evidence & SEAL Pack.

Identity C — pillars + intelligence engines

WORM; FHIR; Advanced Provenance; AKAC; Drug Effectiveness Intelligence; Safety/AE Intelligence; Comparative Intelligence/NBI; National Population Intelligence; QA Gates v2; Anti-bias pack.

6. GOLDEN CAPABILITY LEDGER

Total ledger rows: 62

Each row is a distinct reactivation checklist item. Future teams must not “complete Phase 4” while leaving these unchecked.

P4-A-01 — Data Vocabulary & Mapping Layer

- **PHASE4_CAPABILITY_ID:** P4-A-01
- **CAPABILITY_NAME:** Data Vocabulary & Mapping Layer
- **CAPABILITY_DOMAIN:** Terminology / Interoperability
- **ORIGINAL_PURPOSE:** Canonical drug/therapy and AE vocabulary for cross-site analytics
- **WHAT_IT_DOES:** Builds/maintains RxNorm, ATC/DDD, MedDRA vocabularies; GLP-1 alias/brand-generic mapping; normalises names/spellings/local terms to canonical form; OMOP CDM compatibility
- **PRIMARY_USERS_OR_CONSUMERS:** Ministries, universities, research pipelines, all analytics modules
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.1
- **RELATED_ENGINE_OR_MODULE:** Vocabulary & Mapping
- **INPUTS:** Raw drug/AE strings from Phase 3 and sources
- **OUTPUTS:** Canonical codes + alias tables
- **DEPENDENCIES:** Phase 3 read-only data
- **PRIVACY_OR_SECURITY_CONTROLS:** Feeds privacy-safe aggregates
- **AUDIT_OR_PROVENANCE_CONTROLS:** Supports reproducible coding
- **RESILIENCE_CONTROLS:** N/A
- **LANGUAGE_SUPPORT:** Multilingual input terms
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (suite includes interoperability)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC interoperability themes
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.1; Snapshot module 1

- ELISENCE Phase 4 — Golden Capability & Architecture Reference

Internal Golden Operational Reference
- **KNOWN_LIMITATIONS:** Does not itself diagnose
 - **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Not currently documented as mounted
 - **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED
(default future states start at NOT_YET_REDISCOVERED unless code mission updates)
 - **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — academic research + interoperability evidence

P4-A-02 — Utilization Aggregation Engine

- **PHASE4_CAPABILITY_ID:** P4-A-02
- **CAPABILITY_NAME:** Utilization Aggregation Engine
- **CAPABILITY_DOMAIN:** Utilization Analytics
- **ORIGINAL_PURPOSE:** Population drug-utilization KPIs without hitting raw transactional DBs
- **WHAT_IT_DOES:** Daily/weekly/monthly (and 30-day) aggregated utilisation KPIs per drug/class/cohort (age band, region); pre-computed aggregates
- **PRIMARY_USERS_OR_CONSUMERS:** Ministries, hospitals, Global Insights dashboards
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.2
- **RELATED_ENGINE_OR_MODULE:** Utilization Aggregation
- **INPUTS:** Canonical medication events
- **OUTPUTS:** Utilization KPIs + trend series
- **DEPENDENCIES:** P4-A-01
- **PRIVACY_OR_SECURITY_CONTROLS:** Subject to Privacy Guard before public release
- **AUDIT_OR_PROVENANCE_CONTROLS:** KPI publications WORM-logged
- **RESILIENCE_CONTROLS:** Can Serve-Last-Good snapshots
- **LANGUAGE_SUPPORT:** Dashboard/API consumers multilingual via connectors/PDF
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC KPI themes
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.2; Snapshot module 2
- **KNOWN_LIMITATIONS:** Aggregate descriptive only
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Not documented as mounted
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ISOLATED_TESTED → MOUNTED → INSTITUTION_VISIBLE (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — health-system analytics / public health utilization

P4-A-03 — Effectiveness Engine (Δ Weight 30/60/90)

- **PHASE4_CAPABILITY_ID:** P4-A-03
- **CAPABILITY_NAME:** Effectiveness Engine (Δ Weight 30/60/90)
- **CAPABILITY_DOMAIN:** Effectiveness Analytics
- **ORIGINAL_PURPOSE:** Population-level real-world effectiveness trends for therapies (e.g. GLP-1)
- **WHAT_IT_DOES:** Computes weight-change KPIs at 30/60/90 days when weight data available; architecture allows later activation without redesign
- **PRIMARY_USERS_OR_CONSUMERS:** Ministries, researchers
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.3
- **RELATED_ENGINE_OR_MODULE:** Effectiveness Engine
- **INPUTS:** Weight + therapy cohorts
- **OUTPUTS:** Δ weight KPI profiles by therapy
- **DEPENDENCIES:** P4-A-01; weight data availability
- **PRIVACY_OR_SECURITY_CONTROLS:** K≥50 / DP before public

- **AUDIT_OR_PROVENANCE_CONTROLS:** KPI publish WORM
- **ELISENCE Phase 4 — Golden Capability & Architecture Reference** Internal Golden Operational Reference
- **RESILIENCE_CONTROLS:** Rest of Phase 4 works if inactive
- **LANGUAGE_SUPPORT:** Exportable via PDF engine
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (conditional activation)
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (suite-level)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC KPI themes
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4- خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.3; Snapshot module 3; خلاصه module 3
- **KNOWN_LIMITATIONS:** Requires weight data; not individual clinical recommendation
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Not documented as mounted
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED (weight) → FEATURE_GATED → ENABLED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — clinical evaluation / population effectiveness research (non-diagnostic claims only)

P4-A-04 — Satisfaction Aggregation

- **PHASE4_CAPABILITY_ID:** P4-A-04
- **CAPABILITY_NAME:** Satisfaction Aggregation
- **CAPABILITY_DOMAIN:** Satisfaction Analytics
- **ORIGINAL_PURPOSE:** Population satisfaction signals for therapies/programmes
- **WHAT_IT_DOES:** Builds aggregate patient-reported satisfaction scores
- **PRIMARY_USERS_OR_CONSUMERS:** Ministries, hospitals, payers
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.4
- **RELATED_ENGINE_OR_MODULE:** Satisfaction Aggregation
- **INPUTS:** PRO / satisfaction signals
- **OUTPUTS:** Aggregate satisfaction KPIs
- **DEPENDENCIES:** P4-A-01
- **PRIVACY_OR_SECURITY_CONTROLS:** Privacy Guard
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM on publish
- **RESILIENCE_CONTROLS:** SLG possible
- **LANGUAGE_SUPPORT:** PDF/i18n
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR per-AC
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4- خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.4
- **KNOWN_LIMITATIONS:** Descriptive aggregate only
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Not documented as mounted
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → MOUNTED → INSTITUTION_VISIBLE (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** MEDIUM-HIGH — evaluation / PRO research

P4-A-05 — Adherence Aggregation

- **PHASE4_CAPABILITY_ID:** P4-A-05
- **CAPABILITY_NAME:** Adherence Aggregation
- **CAPABILITY_DOMAIN:** Adherence Analytics
- **ORIGINAL_PURPOSE:** Population adherence pattern insight 8 / 49

- **WHAT_IT_DOES:** Builds adherence KPIs from schedule vs intake signals
- **PRIMARY_USERS_OR_CONSUMERS:** Ministries, hospitals, payers
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.4
- **RELATED_ENGINE_OR_MODULE:** Adherence Aggregation
- **INPUTS:** Schedule/intake signals
- **OUTPUTS:** Aggregate adherence KPIs
- **DEPENDENCIES:** P4-A-01
- **PRIVACY_OR_SECURITY_CONTROLS:** Privacy Guard
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM
- **RESILIENCE_CONTROLS:** SLG
- **LANGUAGE_SUPPORT:** PDF/i18n
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR per-AC
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.4
- **KNOWN_LIMITATIONS:** Not individual clinical adherence coaching claim
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Not documented as mounted
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → MOUNTED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — chronic therapy / GLP-1 programme evaluation

P4-A-06 — Adverse Event (AE) Radar Aggregation

- **PHASE4_CAPABILITY_ID:** P4-A-06
- **CAPABILITY_NAME:** Adverse Event (AE) Radar Aggregation
- **CAPABILITY_DOMAIN:** Safety / Adverse-Event Analytics
- **ORIGINAL_PURPOSE:** Population AE profile comparison across therapies
- **WHAT_IT_DOES:** MedDRA-based population AE radar profiles; descriptive aggregate
- **PRIMARY_USERS_OR_CONSUMERS:** Ministries, hospitals, regulators (oversight)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.4
- **RELATED_ENGINE_OR_MODULE:** AE Radar
- **INPUTS:** AE signals + MedDRA mapping
- **OUTPUTS:** AE radar profiles/KPIs
- **DEPENDENCIES:** P4-A-01 (MedDRA)
- **PRIVACY_OR_SECURITY_CONTROLS:** Privacy Guard
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM
- **RESILIENCE_CONTROLS:** SLG
- **LANGUAGE_SUPPORT:** PDF/i18n
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR per-AC
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.4
- **KNOWN_LIMITATIONS:** Non-diagnostic; not PV system certification
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Not documented as mounted

PHASE4_CAPABILITY_ID: P4-A-07

- CAPABILITY_NAME:** Privacy Guard (k-Anonymity + Differential Privacy)

- **CAPABILITY_DOMAIN:** Privacy

- **ORIGINAL_PURPOSE:** Prevent re-identification from public analytics

- **WHAT_IT_DOES:** Enforces $K \geq 50$ on public cohorts; returns `insufficient_data` if below; DP noise where required; marks `k_anonymity/dp_applied`

- **PRIMARY_USERS_OR_CONSUMERS:** All public/research consumers

- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.5

- **RELATED_ENGINE_OR_MODULE:** Privacy Guard

- **INPUTS:** Candidate aggregate responses

- **OUTPUTS:** Privacy-enforced responses or insufficient_data

- **DEPENDENCIES:** Cohort builders

- **PRIVACY_OR_SECURITY_CONTROLS:** Core control

- **AUDIT_OR_PROVENANCE_CONTROLS:** Enforcement auditable

- **RESILIENCE_CONTROLS:** Fails closed to insufficient_data

- **LANGUAGE_SUPPORT:** N/A

- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED

- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (privacy & k-anon tests)

- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC privacy themes

- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR

- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR

- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf

- **SOURCE_SECTIONS:** Phase4-Report §2.5; Snapshot module 5

- **KNOWN_LIMITATIONS:** Absolute “never leaks” marketing language elsewhere must not be overstated

- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Control plane not user-visible

- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ISOLATED_TESTED → INTEGRATION_TESTED → ENABLED
(default future states start at NOT_YET_REDISCOVERED unless code mission updates)

- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — privacy-preserving analytics collaboration prerequisite

PHASE4_CAPABILITY_ID: P4-A-08

- **CAPABILITY_NAME:** RBAC Role Model (Public / Research / Admin)

- **CAPABILITY_DOMAIN:** Zero Trust / RBAC

- **ORIGINAL_PURPOSE:** Bound API access by role

- **WHAT_IT_DOES:** Public=conservative aggregates; Research=more detailed privacy-safe aggregates with API keys; Admin=WORM/config/export; minimum role per route

- **PRIMARY_USERS_OR_CONSUMERS:** Public dashboards, universities, ministry admins

- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.6

- **RELATED_ENGINE_OR_MODULE:** RBAC

- **INPUTS:** Auth identity/API key

- **OUTPUTS:** Authorized responses or denial

- **DEPENDENCIES:** Auth provider

- **PRIVACY_OR_SECURITY_CONTROLS:** Role bounds data detail

- **AUDIT_OR_PROVENANCE_CONTROLS:** Admin actions WORM

- **RESILIENCE_CONTROLS:** Deny by default posture

- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (Admin/WORM/RBAC checks)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC RBAC/security
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.6
- **KNOWN_LIMITATIONS:** Must map to modern auth model
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Admin/research surfaces
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → MOUNTED → FEATURE_GATED
(default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — university research access vs public dashboards

P4-A-09 — Global Insights UI Connectors

- **PHASE4_CAPABILITY_ID:** P4-A-09
- **CAPABILITY_NAME:** Global Insights UI Connectors
- **CAPABILITY_DOMAIN:** Connectors / Global Insights
- **ORIGINAL_PURPOSE:** Expose analytics to dashboards without write-back
- **WHAT_IT_DOES:** Read-only APIs for Global Insights tab: ≥3 charts (e.g. utilization drug-vs-drug, AE radar, trends) + ≥3 KPI cards; consumable by hospital BI / ministry / academic portals
- **PRIMARY_USERS_OR_CONSUMERS:** Hospital BI, ministry dashboards, academic portals
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.7
- **RELATED_ENGINE_OR_MODULE:** Global Insights Connectors
- **INPUTS:** Privacy-enforced KPIs
- **OUTPUTS:** Chart/KPI API payloads
- **DEPENDENCIES:** P4-A-02..06, P4-A-07, P4-A-08
- **PRIVACY_OR_SECURITY_CONTROLS:** Read-only + Privacy Guard
- **AUDIT_OR_PROVENANCE_CONTROLS:** Access/export logging
- **RESILIENCE_CONTROLS:** Can serve last-good
- **LANGUAGE_SUPPORT:** UI may localize via consumers/PDF
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (suite-level)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR per-AC
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.7; Snapshot module 7
- **KNOWN_LIMITATIONS:** Connector design; consumer UI may be separate
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended dashboard surface
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → MOUNTED → USER_VISIBLE/INSTITUTION_VISIBLE (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — demonstrable institutional analytics surface

P4-A-10 — Monitoring & Metrics Export

- **PHASE4_CAPABILITY_ID:** P4-A-10
- **CAPABILITY_NAME:** Monitoring & Metrics Export
- **CAPABILITY_DOMAIN:** Observability / Monitoring
- **ORIGINAL_PURPOSE:** Operational observability for national-scale use
- **WHAT_IT_DOES:** Exposes latency/error/throughput metrics for stacks such as Prometheus/Grafana

- **PRIMARY_USERS_OR_CONSUMERS:** Ops, admins, institutional IT
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.8
- **RELATED_ENGINE_OR_MODULE:** Monitoring
- **INPUTS:** Runtime telemetry
- **OUTPUTS:** Metrics series
- **DEPENDENCIES:** Runtime
- **PRIVACY_OR_SECURITY_CONTROLS:** Avoid PII in metrics
- **AUDIT_OR_PROVENANCE_CONTROLS:** Complements WORM
- **RESILIENCE_CONTROLS:** Supports detection of failure modes
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (observability themes)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC observability
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.8
- **KNOWN_LIMITATIONS:** Stack choice may differ in modern runtime
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Ops-only
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ENABLED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** MEDIUM — operational maturity evidence

P4-A-11 — WORM Ledger (Analytics)

- **PHASE4_CAPABILITY_ID:** P4-A-11
- **CAPABILITY_NAME:** WORM Ledger (Analytics)
- **CAPABILITY_DOMAIN:** Audit / Provenance / WORM
- **ORIGINAL_PURPOSE:** Immutable record of critical analytics actions
- **WHAT_IT_DOES:** Append-only WORM for exports, KPI publications, config changes, incident events
- **PRIMARY_USERS_OR_CONSUMERS:** Regulators, auditors, admins, partner universities
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.8
- **RELATED_ENGINE_OR_MODULE:** WORM Ledger
- **INPUTS:** Critical action events
- **OUTPUTS:** Immutable audit records
- **DEPENDENCIES:** Storage
- **PRIVACY_OR_SECURITY_CONTROLS:** Should not store unnecessary PII
- **AUDIT_OR_PROVENANCE_CONTROLS:** Core
- **RESILIENCE_CONTROLS:** Append-only integrity
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (WORM checks)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC WORM/audit
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.8
- **KNOWN_LIMITATIONS:** Implementation medium not fully specified
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Admin/auditor
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ISOLATED_TESTED → ENABLED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)

P4-A-12 — Quality Gates / Data Contracts / JSON Schema Versioning

- **PHASE4_CAPABILITY_ID:** P4–A–12
- **CAPABILITY_NAME:** Quality Gates / Data Contracts / JSON Schema Versioning
- **CAPABILITY_DOMAIN:** Data Contracts / Schema / Versioning
- **ORIGINAL_PURPOSE:** Predictable validated analytics outputs
- **WHAT_IT_DOES:** Versioned JSON Schemas for KPIs/payloads; validates schema_version and kpi_version on responses
- **PRIMARY_USERS_OR_CONSUMERS:** Integrators, QA, universities
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.9
- **RELATED_ENGINE_OR_MODULE:** Quality Gates
- **INPUTS:** Candidate payloads
- **OUTPUTS:** Validated contracted responses
- **DEPENDENCIES:** Schema registry
- **PRIVACY_OR_SECURITY_CONTROLS:** Contracts can encode privacy fields
- **AUDIT_OR_PROVENANCE_CONTROLS:** Version evidence
- **RESILIENCE_CONTROLS:** Reject invalid payloads
- **LANGUAGE_SUPPORT:** Schema field localization separate
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (schema validation)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC versioning
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.9; Snapshot module 9
- **KNOWN_LIMITATIONS:** Exact schema catalogue not listed in docs
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Integrator-facing
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → INTEGRATION_TESTED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — reproducible research pipelines

P4-A-13 — Synthetic Data Generators

- **PHASE4_CAPABILITY_ID:** P4–A–13
- **CAPABILITY_NAME:** Synthetic Data Generators
- **CAPABILITY_DOMAIN:** Synthetic Data / Test Infrastructure
- **ORIGINAL_PURPOSE:** Safe reproducible QA without real PHI
- **WHAT_IT_DOES:** Generates synthetic datasets for testing and demos
- **PRIMARY_USERS_OR_CONSUMERS:** QA, engineers, demos to ministries/universities
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.9
- **RELATED_ENGINE_OR_MODULE:** Synthetic Data
- **INPUTS:** Schema/contracts
- **OUTPUTS:** Synthetic datasets
- **DEPENDENCIES:** P4-A-12
- **PRIVACY_OR_SECURITY_CONTROLS:** Avoids real data
- **AUDIT_OR_PROVENANCE_CONTROLS:** Test evidence packs
- **RESILIENCE_CONTROLS:** N/A
- **LANGUAGE_SUPPORT:** Can seed multilingual strings
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (synthetic & schema validation suite)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR

- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.9; خلاصه module 9
- **KNOWN_LIMITATIONS:** Not a substitute for real-world validation studies
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Internal/QA
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ENABLED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** MEDIUM — safe demo/reproducibility

P4-A-14 — Circuit Breaker (Phase 3 upstream)

- **PHASE4_CAPABILITY_ID:** P4-A-14
- **CAPABILITY_NAME:** Circuit Breaker (Phase 3 upstream)
- **CAPABILITY_DOMAIN:** Resilience / Failure Safety
- **ORIGINAL_PURPOSE:** Protect analytics when upstream Phase 3 fails
- **WHAT_IT_DOES:** Circuit breaker for upstream Phase 3 dependencies
- **PRIMARY_USERS_OR_CONSUMERS:** System continuity
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.10
- **RELATED_ENGINE_OR_MODULE:** Failure Modes Layer
- **INPUTS:** Upstream health
- **OUTPUTS:** Open/half-open/closed breaker states
- **DEPENDENCIES:** Phase 3
- **PRIVACY_OR_SECURITY_CONTROLS:** Prevents unsafe degraded leakage paths
- **AUDIT_OR_PROVENANCE_CONTROLS:** Incident events WORM
- **RESILIENCE_CONTROLS:** Core
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (failure mode simulations)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.10
- **KNOWN_LIMITATIONS:** Thresholds not detailed in docs
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Ops
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ISOLATED_TESTED → ENABLED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — resilience evidence

P4-A-15 — Serve-Last-Good (SLG) Mode

- **PHASE4_CAPABILITY_ID:** P4-A-15
- **CAPABILITY_NAME:** Serve-Last-Good (SLG) Mode
- **CAPABILITY_DOMAIN:** Resilience / Failure Safety
- **ORIGINAL_PURPOSE:** Continue serving safe snapshots during failures
- **WHAT_IT_DOES:** Serve last-good aggregates when live recompute/upstream unavailable
- **PRIMARY_USERS_OR_CONSUMERS:** Dashboard/API consumers during incidents
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.10
- **RELATED_ENGINE_OR_MODULE:** Failure Modes Layer
- **INPUTS:** Cached last-good snapshots
- **OUTPUTS:** Safe historical aggregates

- **CAPABILITY_DOMAIN:** Resilience / Failure Safety
- **ORIGINAL_PURPOSE:** Recover analytics after outages
- **WHAT_IT_DOES:** Backfill and recovery routines after source/network/partial failures
- **PRIMARY_USERS_OR_CONSUMERS:** Ops/admin
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.10
- **RELATED_ENGINE_OR_MODULE:** Failure Modes Layer
- **INPUTS:** Missed windows / failed jobs
- **OUTPUTS:** Rebuilt aggregates
- **DEPENDENCIES:** ETL pipelines
- **PRIVACY_OR_SECURITY_CONTROLS:** Must re-apply Privacy Guard
- **AUDIT_OR_PROVENANCE_CONTROLS:** Recovery events WORM
- **RESILIENCE_CONTROLS:** Core
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (failure simulations)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4- خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.10
- **KNOWN_LIMITATIONS:** Procedures not fully enumerated
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Ops
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ISOLATED_TESTED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** MEDIUM

P4-A-18 — Acceptance Criteria Engine (AC-1..AC-20)

- **PHASE4_CAPABILITY_ID:** P4-A-18
- **CAPABILITY_NAME:** Acceptance Criteria Engine (AC-1..AC-20)
- **CAPABILITY_DOMAIN:** Acceptance / Go-No-Go
- **ORIGINAL_PURPOSE:** Machine-checkable acceptance of Phase 4 readiness
- **WHAT_IT_DOES:** Encodes 20 ACs: isolation; RxNorm/ATC/MedDRA/OMOP interoperability; API latency p95/p99; ETL performance/scalability; k-anon & DP; RBAC & security; WORM & audit; retention & erasure; observability; safe merge/deployment path; etc.
- **PRIMARY_USERS_OR_CONSUMERS:** Admins, health authorities, ethics boards, university partners
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.11
- **RELATED_ENGINE_OR_MODULE:** AC Engine
- **INPUTS:** System under test
- **OUTPUTS:** Per-AC pass/fail + evidence refs
- **DEPENDENCIES:** Test harness / pytest
- **PRIVACY_OR_SECURITY_CONTROLS:** Includes privacy ACs
- **AUDIT_OR_PROVENANCE_CONTROLS:** Evidence references
- **RESILIENCE_CONTROLS:** Includes failure/merge safety themes
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (AC tests in ~244 suite)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED; final GO result not quoted in folder
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4- خلاصه مدیریتی.pdf

- **SOURCE_SECTIONS:** Phase4-Report §2.11 pp6-7; Snapshot module 11
- **KNOWN_LIMITATIONS:** Individual AC IDs list themes not full AC text table
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Admin
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ISOLATED_TESTED → PRODUCTION_VALIDATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — evidence of machine acceptance discipline

P4-A-19 — Go/No-Go Harness Endpoint

- **PHASE4_CAPABILITY_ID:** P4-A-19
- **CAPABILITY_NAME:** Go/No-Go Harness Endpoint
- **CAPABILITY_DOMAIN:** Acceptance / Go-No-Go
- **ORIGINAL_PURPOSE:** Pre-deployment self-certification gate
- **WHAT_IT_DOES:** Admin-only endpoint returns overall_status GO/NO-GO, each AC pass/fail, evidence references
- **PRIMARY_USERS_OR_CONSUMERS:** Admins, deployment authorities
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.11
- **RELATED_ENGINE_OR_MODULE:** Go/No-Go Harness
- **INPUTS:** AC Engine results
- **OUTPUTS:** GO/NO-GO JSON
- **DEPENDENCIES:** P4-A-18
- **PRIVACY_OR_SECURITY_CONTROLS:** Admin-only
- **AUDIT_OR_PROVENANCE_CONTROLS:** Invocation should be logged
- **RESILIENCE_CONTROLS:** NO-GO blocks unsafe deploy intent
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (suite-level)
- **REPORTED_ACCEPTANCE_STATE:** Harness documented; recorded GO outcome NOT FOUND in folder
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisenca — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §2.11
- **KNOWN_LIMITATIONS:** No stored GO certificate in documentary set
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Admin
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ENABLED → PRODUCTION_VALIDATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH — institutional confidence mechanism

P4-A-20 — Multi-Language PDF Export Engine (EN/FA/AR/TR/RO)

- **PHASE4_CAPABILITY_ID:** P4-A-20
- **CAPABILITY_NAME:** Multi-Language PDF Export Engine (EN/FA/AR/TR/RO)
- **CAPABILITY_DOMAIN:** Reporting / Export / Multilingual
- **ORIGINAL_PURPOSE:** Official multilingual evidence packs
- **WHAT_IT_DOES:** Generates official PDF reports in EN/FA/AR(RTL)/TR/RO; correct RTL/typography; non-PII aggregate-only; WORM-ready; voice/sensor metadata support noted in Persian snapshot
- **PRIMARY_USERS_OR_CONSUMERS:** Ministries, hospitals, universities, funders
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine / Module 2.12
- **RELATED_ENGINE_OR_MODULE:** PDF Export Engine
- **INPUTS:** Aggregate analytics + templates
- **OUTPUTS:** Locked PDF evidence packs
- **DEPENDENCIES:** P4-A-02..06, P4-A-07, P4-A-11
- **PRIVACY_OR_SECURITY_CONTROLS:** Non-PII only

- AUDIT_OR_PROVENANCE_CONTROLS:** Export WORM
- ELISENCE Phase 4 — Golden Capability & Architecture Reference**
- RESILIENCE_CONTROLS:** N/A
- LANGUAGE_SUPPORT:** EN FA AR TR RO + RTL
- REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (language themes referenced programme-wide)
- REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR per-AC
- REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4- خلاصه مدیریتی.pdf
- SOURCE_SECTIONS:** Phase4-Report §2.12; Snapshot module 12; خلاصه module 12
- KNOWN_LIMITATIONS:** Not clinical case report generator
- CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Institution/research export
- FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → ISOLATED_TESTED → INSTITUTION_VISIBLE/RESEARCH_VISIBLE
(default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- ALBANIA_EVIDENCE_RELEVANCE:** HIGH — multilingual collaboration evidence (Albania/EIT dossiers)

P4-A-21 — Phase 3 Isolation / Read-Only Boundary

- PHASE4_CAPABILITY_ID:** P4-A-21
- CAPABILITY_NAME:** Phase 3 Isolation / Read-Only Boundary
- CAPABILITY_DOMAIN:** Isolation / Trust Boundary
- ORIGINAL_PURPOSE:** Keep operational care layer transactional and uncorrupted
- WHAT_IT_DOES:** Phase 4 sits on top of Phase 3; never writes back; Phase 3 fully isolated in status claims
- PRIMARY_USERS_OR_CONSUMERS:** Architecture / safety
- ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine cross-cutting
- RELATED_ENGINE_OR_MODULE:** Isolation boundary
- INPUTS:** Phase 3 read paths
- OUTPUTS:** Aggregates only
- DEPENDENCIES:** Phase 3
- PRIVACY_OR_SECURITY_CONTROLS:** Reduces blast radius
- AUDIT_OR_PROVENANCE_CONTROLS:** Isolation is AC theme
- RESILIENCE_CONTROLS:** Supports failure isolation
- LANGUAGE_SUPPORT:** N/A
- REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (isolation AC)
- REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC-1 isolation theme
- REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4- خلاصه مدیریتی.pdf
- SOURCE_SECTIONS:** Phase4-Report §1, §5, status header
- KNOWN_LIMITATIONS:** Must be re-verified in modern runtime
- CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Architectural
- FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → INTEGRATION_TESTED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- ALBANIA_EVIDENCE_RELEVANCE:** HIGH — safe layering narrative

P4-A-22 — Zero-Trust Security Posture (Analytics)

- PHASE4_CAPABILITY_ID:** P4-A-22
- CAPABILITY_NAME:** Zero-Trust Security Posture (Analytics)
- CAPABILITY_DOMAIN:** Zero Trust / RBAC
- ORIGINAL_PURPOSE:** No implicit trust in analytics exposure paths

- **WHAT_IT_DOES:** Programme built under Zero-Trust security; Privacy Guard + RBAC as enforcement points
- **PRIMARY_USERS_OR_CONSUMERS:** All
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Cross-cutting
- **RELATED_ENGINE_OR_MODULE:** Zero-Trust controls
- **INPUTS:** Requests
- **OUTPUTS:** Authorized privacy-safe responses
- **DEPENDENCIES:** P4-A-07, P4-A-08
- **PRIVACY_OR_SECURITY_CONTROLS:** Core
- **AUDIT_OR_PROVENANCE_CONTROLS:** Core
- **RESILIENCE_CONTROLS:** Fail closed
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / claimed IMPLEMENTED
- **REPORTED_TEST_STATE:** DOCUMENTED AS TESTED (security checks)
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTED AS DESIGNED into AC security
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §1
- **KNOWN_LIMITATIONS:** Principle + controls; not a formal ZT certification
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Control plane
- **FUTURE_REACTIVATION_CHECK:** DEPENDENCIES_VALIDATED → INTEGRATION_TESTED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH

P4-A-23 — Multi-Channel Data Ingestion Posture (sensor/voice/manual)

- **PHASE4_CAPABILITY_ID:** P4-A-23
- **CAPABILITY_NAME:** Multi-Channel Data Ingestion Posture (sensor/voice/manual)
- **CAPABILITY_DOMAIN:** Data Architecture
- **ORIGINAL_PURPOSE:** Accept multi-channel inputs into analytics layer
- **WHAT_IT_DOES:** Designed for sensor, voice, and manual inputs feeding analytics
- **PRIMARY_USERS_OR_CONSUMERS:** Upstream phases / pipelines
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Analytics Engine cross-cutting
- **RELATED_ENGINE_OR_MODULE:** Ingestion posture
- **INPUTS:** Sensor/voice/manual
- **OUTPUTS:** Normalized analytics inputs
- **DEPENDENCIES:** Upstream emitters
- **PRIVACY_OR_SECURITY_CONTROLS:** Must pass privacy before public
- **AUDIT_OR_PROVENANCE_CONTROLS:** Pipeline audit
- **RESILIENCE_CONTROLS:** Circuit breaker to upstream
- **LANGUAGE_SUPPORT:** Voice multilingual implications
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / claimed part of architecture
- **REPORTED_TEST_STATE:** DOCUMENTARY STATUS UNCLEAR at channel granularity
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf
- **SOURCE_SECTIONS:** Phase4-Report §1
- **KNOWN_LIMITATIONS:** Depth of connectors not fully specified in Analytics docs
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Backend

PHASE4_CAPABILITY_ID: P4-A-24

CAPABILITY_NAME: BBVP Discipline (Build→Back→Verify→Prove)

CAPABILITY_DOMAIN: Programme / Quality Method

ORIGINAL_PURPOSE: Evidence-oriented engineering method

WHAT_IT_DOES: Programme states BBVP as governing method; pytest auditor replay intended

PRIMARY_USERS_OR_CONSUMERS: Engineering / auditors

ARCHITECTURAL_OWNER_OR_SUBSYSTEM: Programme

RELATED_ENGINE_OR_MODULE: BBVP

INPUTS: Code+tests+evidence

OUTPUTS: Provable verification artefacts

DEPENDENCIES: Test suites

PRIVACY_OR_SECURITY_CONTROLS: N/A

AUDIT_OR_PROVENANCE_CONTROLS: Prove step

RESILIENCE_CONTROLS: N/A

LANGUAGE_SUPPORT: N/A

- SOURCE DOCUMENTS:** Phase4-Report.pdf; Elisence — Phase 4 (Global Analytics Engine).pdf; Phase4-خلاصه مدیریتی.pdf

PHASE4_CAPABILITY_ID: P4-G-01

CAPABILITY_NAME: GHX Foundation Layer (Identity & Jurisdiction Engine)

CAPABILITY_DOMAIN: Foundation

ORIGINAL_PURPOSE: Pseudonymous IDs + jurisdiction profiles (UK_NHS, EU_GDPR, GCC_PDPL, US_HIPAA...)

WHAT_IT_DOES: Pseudonymous IDs + jurisdiction profiles (UK_NHS, EU_GDPR, GCC_PDPL, US_HIPAA...); decides legal rule-set per export

PRIMARY_USERS_OR_CONSUMERS: See GHX actors/modes and product contracts

ARCHITECTURAL_OWNER_OR_SUBSYSTEM: Tier 0

RELATED_ENGINE_OR_MODULE: GHX Foundation Layer

INPUTS: Documented per component

OUTPUTS: Documented per component

DEPENDENCIES: GHX tier ordering

PRIVACY_OR_SECURITY_CONTROLS: Consent/AKAC/DP/mode as applicable

AUDIT_OR_PROVENANCE_CONTROLS: WORM/ledger as applicable

RESILIENCE_CONTROLS: Validator block / OFF gate

LANGUAGE_SUPPORT: EN/FA/AR/TR/RO + RTL where applicable

- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (Tier-0 models & tests 100% green per سندGHX)
- **REPORTED_TEST_STATE:** See ELISENCE-Phase 4 - Golden Capability & Architecture Reference GHX tier statuses / GHX. Report 164/164 claim (conflicts) Internal Golden Operational Reference
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-02 — Locale & Culture Engine

- **PHASE4_CAPABILITY_ID:** P4-G-02
- **CAPABILITY_NAME:** Locale & Culture Engine
- **CAPABILITY_DOMAIN:** Multilingual / Cultural
- **ORIGINAL_PURPOSE:** EN/FA/AR/TR/RO + RTL
- **WHAT_IT_DOES:** EN/FA/AR/TR/RO + RTL; cultural baselines BMI/fasting/heat
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 0
- **RELATED_ENGINE_OR_MODULE:** Locale & Culture Engine
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (cultural models & tests per سندGHX)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-03 — Global Policy Registry

- **PHASE4_CAPABILITY_ID:** P4-G-03
- **CAPABILITY_NAME:** Global Policy Registry
- **CAPABILITY_DOMAIN:** Regulatory-Aware Policy
- **ORIGINAL_PURPOSE:** Versioned NHS/FDA/MDR/GCC/local policy packs for CDS/consent/export
- **WHAT_IT_DOES:** Versioned NHS/FDA/MDR/GCC/local policy packs for CDS/consent/export
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 0

- **RELATED_ENGINE_OR_MODULE:** Global Policy Registry
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (with Tier-0)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-04 — ISecureDataEmitter Interface

- **PHASE4_CAPABILITY_ID:** P4-G-04
- **CAPABILITY_NAME:** ISecureDataEmitter Interface
- **CAPABILITY_DOMAIN:** Integration Contract
- **ORIGINAL_PURPOSE:** Mandatory outbound interface for every phase/engine into GHX
- **WHAT_IT_DOES:** Mandatory outbound interface for every phase/engine into GHX
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 1 / Interface
- **RELATED_ENGINE_OR_MODULE:** ISecureDataEmitter Interface
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (emitter/pipeline skeleton per سندGHX)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

- **PHASE4_CAPABILITY_ID:** P4-G-05
- **CAPABILITY_NAME:** SecureWrapper
- **CAPABILITY_DOMAIN:** Governance Gatekeeper
- **ORIGINAL_PURPOSE:** Consent + jurisdiction + privacy + mode checks before FHIR/WORM
- **WHAT_IT_DOES:** Consent + jurisdiction + privacy + mode checks before FHIR/WORM
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 1
- **RELATED_ENGINE_OR_MODULE:** SecureWrapper
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (skeleton/path documented)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-06 — FHIRTransformer

- **PHASE4_CAPABILITY_ID:** P4-G-06
- **CAPABILITY_NAME:** FHIRTransformer
- **CAPABILITY_DOMAIN:** Interoperability
- **ORIGINAL_PURPOSE:** Maps payloads to FHIR Observation/Condition/Meds/Encounter + SNOMED/LOINC/RxNorm/ATC
- **WHAT_IT_DOES:** Maps payloads to FHIR Observation/Condition/Meds/Encounter + SNOMED/LOINC/RxNorm/ATC
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 1
- **RELATED_ENGINE_OR_MODULE:** FHIRTransformer
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (stub/tests green)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder

- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-07 — FHIRValidator

- **PHASE4_CAPABILITY_ID:** P4-G-07
- **CAPABILITY_NAME:** FHIRValidator
- **CAPABILITY_DOMAIN:** Interoperability / Safety
- **ORIGINAL_PURPOSE:** Validates UK Core/US Core/International profiles
- **WHAT_IT_DOES:** Validates UK Core/US Core/International profiles; blocks non-conformant
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 1
- **RELATED_ENGINE_OR_MODULE:** FHIRValidator
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (validator tests green)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-08 — ConsentEngine & Smart-Contract Consent Schema

- **PHASE4_CAPABILITY_ID:** P4-G-08
- **CAPABILITY_NAME:** ConsentEngine & Smart-Contract Consent Schema
- **CAPABILITY_DOMAIN:** Consent / Authorization
- **ORIGINAL_PURPOSE:** Consent objects
- **WHAT_IT_DOES:** Consent objects; ephemeral tokens; kill-switch revocation
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 2
- **RELATED_ENGINE_OR_MODULE:** ConsentEngine & Smart-Contract Consent Schema
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable

- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **ELISENCE Phase 4 – Golden Capability & Architecture Reference** Internal Golden Operational Reference
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (consent core present; full scenarios later)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-09 — Privacy & Anonymisation Engine (AKAC + DP Manager)

- **PHASE4_CAPABILITY_ID:** P4-G-09
- **CAPABILITY_NAME:** Privacy & Anonymisation Engine (AKAC + DP Manager)
- **CAPABILITY_DOMAIN:** Privacy
- **ORIGINAL_PURPOSE:** Adaptive k-anonymity + DP budgeting for ministry/research exports
- **WHAT_IT_DOES:** Adaptive k-anonymity + DP budgeting for ministry/research exports
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 2
- **RELATED_ENGINE_OR_MODULE:** Privacy & Anonymisation Engine
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / PARTIAL (policy structures present; budgeting in M4 plan)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-10 — Actors & Modes Contract Engine

- **PHASE4_CAPABILITY_ID:** P4-G-10
- **CAPABILITY_NAME:** Actors & Modes Contract Engine
- **CAPABILITY_DOMAIN:** RBAC/ABAC

- **ORIGINAL_PURPOSE:** Actors citizen/clinician/hospital_admin/ministry_researcher/payer; modes Citizen/Clinician/Ministry/Research; explicit error codes
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Cross-cutting
- **RELATED_ENGINE_OR_MODULE:** Actors & Modes Contract Engine
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (fully covered by tests per سندGHX)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-11 — CulturalBaseline Profiles & GHX i18n Registry

- **PHASE4_CAPABILITY_ID:** P4-G-11
- **CAPABILITY_NAME:** CulturalBaseline Profiles & GHX i18n Registry
- **CAPABILITY_DOMAIN:** Multilingual / Cultural
- **ORIGINAL_PURPOSE:** Region profiles UK/QA/TR/IR/RO...
- **WHAT_IT_DOES:** Region profiles UK/QA/TR/IR/RO...; central i18n keys for messages/CDS/dashboards; FA/AR RTL
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Cross-cutting
- **RELATED_ENGINE_OR_MODULE:** CulturalBaseline Profiles & GHX i18n Registry
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers

- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-12 — WORM / Ledger & Provenance Chain (GHX)

- **PHASE4_CAPABILITY_ID:** P4-G-12
- **CAPABILITY_NAME:** WORM / Ledger & Provenance Chain (GHX)
- **CAPABILITY_DOMAIN:** Audit / Provenance
- **ORIGINAL_PURPOSE:** Append-only hash-chained ledger for exports/policy decisions
- **WHAT_IT_DOES:** Append-only hash-chained ledger for exports/policy decisions
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Cross-cutting
- **RELATED_ENGINE_OR_MODULE:** WORM / Ledger & Provenance Chain
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS IMPLEMENTED (ledger in Tier-1 path)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-13 — GHXQueue & Governance Routing

- **PHASE4_CAPABILITY_ID:** P4-G-13
- **CAPABILITY_NAME:** GHXQueue & Governance Routing
- **CAPABILITY_DOMAIN:** Data Flow
- **ORIGINAL_PURPOSE:** Secure queue staging validated FHIR for warehouse/analytics/CDS/export
- **WHAT_IT_DOES:** Secure queue staging validated FHIR for warehouse/analytics/CDS/export; jurisdiction routing
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 1→2/3
- **RELATED_ENGINE_OR_MODULE:** GHXQueue & Governance Routing
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable

- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / PARTIAL
- **REPORTED_TEST_STATE:** See ^{ELISENCE Phase 4 – Golden Capability & Architecture Reference} GHX tier statuses / GHX. Report 164/164 claim (conflicts) ^{Internal Golden Operational Reference}
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-14 — Data Warehouse & Federated Query Contracts

- **PHASE4_CAPABILITY_ID:** P4-G-14
- **CAPABILITY_NAME:** Data Warehouse & Federated Query Contracts
- **CAPABILITY_DOMAIN:** Population / Research Infrastructure
- **ORIGINAL_PURPOSE:** Country-isolated warehouses FHIR/OMOP/risk/KPI
- **WHAT_IT_DOES:** Country-isolated warehouses FHIR/OMOP/risk/KPI; federated query returns aggregates only
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 3
- **RELATED_ENGINE_OR_MODULE:** Data Warehouse & Federated Query Contracts
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (implementation tied to M4–M6)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-15 — Federated Learning Layer

- **PHASE4_CAPABILITY_ID:** P4-G-15
- **CAPABILITY_NAME:** Federated Learning Layer
- **CAPABILITY_DOMAIN:** Research / AI Governance
- **ORIGINAL_PURPOSE:** Local nodes train
- **WHAT_IT_DOES:** Local nodes train; models travel, data does not; secure aggregation PoC
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 3

- **RELATED_ENGINE_OR_MODULE:** Federated Learning Layer
- **ELISENCE Phase 4 – Golden Capability & Architecture Reference**
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (M6 / DONE gate ≥ 2 nodes)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-16 — Digital Twin Orchestration Contract

- **PHASE4_CAPABILITY_ID:** P4–G–16
- **CAPABILITY_NAME:** Digital Twin Orchestration Contract
- **CAPABILITY_DOMAIN:** Health Intelligence / Simulation
- **ORIGINAL_PURPOSE:** APIs for Cardiac, Diabetes, Women+, Kids+, Metabolic twins
- **WHAT_IT_DOES:** APIs for Cardiac, Diabetes, Women+, Kids+, Metabolic twins; non-diagnostic explainable simulations
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 4
- **RELATED_ENGINE_OR_MODULE:** Digital Twin Orchestration Contract
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (contracts locked; endpoints M7–M9)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

- **PHASE4_CAPABILITY_ID:** P4-G-17
- **CAPABILITY_NAME:** CDS Hooks Contract Layer
- **CAPABILITY_DOMAIN:** Clinical Decision Support
- **ORIGINAL_PURPOSE:** HL7 CDS Hooks patient-view/order-select/order-sign/encounter-start
- **WHAT_IT_DOES:** HL7 CDS Hooks patient-view/order-select/order-sign/encounter-start; combines phases 7/8/9/16/17 + environment; 5-lang culture-aware non-diagnostic cards
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 2/4
- **RELATED_ENGINE_OR_MODULE:** CDS Hooks Contract Layer
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / PARTIAL (contracts+tests; full CDS server/UI later)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-18 — Government & Ministry Portal Contract

- **PHASE4_CAPABILITY_ID:** P4-G-18
- **CAPABILITY_NAME:** Government & Ministry Portal Contract
- **CAPABILITY_DOMAIN:** Government / Institutional
- **ORIGINAL_PURPOSE:** Minister dashboards: macro KPIs, heatmaps, foresight
- **WHAT_IT_DOES:** Minister dashboards: macro KPIs, heatmaps, foresight; anonymised tiles with privacy+WORM+culture
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 4 Product A
- **RELATED_ENGINE_OR_MODULE:** Government & Ministry Portal Contract
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (DONE gate: 5-lang+RTL demo-ready)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR

- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx, Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-19 — Hospital & Clinic Integration Contract

- **PHASE4_CAPABILITY_ID:** P4-G-19
- **CAPABILITY_NAME:** Hospital & Clinic Integration Contract
- **CAPABILITY_DOMAIN:** Institutional / Clinical Systems
- **ORIGINAL_PURPOSE:** FHIR API + CDS for EHR (Epic/Cerner-style) and ward dashboards
- **WHAT_IT_DOES:** FHIR API + CDS for EHR (Epic/Cerner-style) and ward dashboards; clinician pseudonymous vs citizen wellness separation
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 4 Product B
- **RELATED_ENGINE_OR_MODULE:** Hospital & Clinic Integration Contract
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-20 — Research & OMOP Export Contract

- **PHASE4_CAPABILITY_ID:** P4-G-20
- **CAPABILITY_NAME:** Research & OMOP Export Contract
- **CAPABILITY_DOMAIN:** Research & Evidence
- **ORIGINAL_PURPOSE:** OMOP CDM, FHIR NDJSON, DP-protected aggregates
- **WHAT_IT_DOES:** OMOP CDM, FHIR NDJSON, DP-protected aggregates; schema/codebooks/consent/WORM evidence packs
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Tier 4 Product C
- **RELATED_ENGINE_OR_MODULE:** Research & OMOP Export Contract
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component

- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-21 — Phase 4 GHX Evidence & SEAL Pack

- **PHASE4_CAPABILITY_ID:** P4-G-21
- **CAPABILITY_NAME:** Phase 4 GHX Evidence & SEAL Pack
- **CAPABILITY_DOMAIN:** Evidence / Seal Infrastructure
- **ORIGINAL_PURPOSE:** Evidence ZIP structure: tests, configs, policy docs, SHA-256 manifest
- **WHAT_IT_DOES:** Evidence ZIP structure: tests, configs, policy docs, SHA-256 manifest; defines DONE review pack
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Programme
- **RELATED_ENGINE_OR_MODULE:** Phase 4 GHX Evidence & SEAL Pack
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (when implemented and sealed)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-22 — Feature-Flag Hard-OFF Gate (GHX BBVP BACK)

- **PHASE4_CAPABILITY_ID:** P4-G-22
- **CAPABILITY_NAME:** Feature-Flag Hard-OFF Gate (GHX BBVP BACK)

- CAPABILITY_DOMAIN:** Feature Gating / Safety
- ORIGINAL_PURPOSE:** OFF = hard 503/denial with no leakage or fake output
- WHAT_IT_DOES:** OFF = hard 503/denial with no leakage or fake output
- PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Cross-cutting BBVP
- RELATED_ENGINE_OR_MODULE:** Feature-Flag Hard-OFF Gate
- INPUTS:** Documented per component
- OUTPUTS:** Documented per component
- DEPENDENCIES:** GHX tier ordering
- PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- RESILIENCE_CONTROLS:** Validator block / OFF gate
- LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / claimed control
- REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-23 — Single Canonical Emission Path

- **PHASE4_CAPABILITY_ID:** P4-G-23
- **CAPABILITY_NAME:** Single Canonical Emission Path
- **CAPABILITY_DOMAIN:** Isolation / Data Flow
- **ORIGINAL_PURPOSE:** No cross-phase DB shortcuts
- **WHAT_IT_DOES:** No cross-phase DB shortcuts; no parallel emitters; only ISecureDataEmitter path
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Architecture non-negotiable
- **RELATED_ENGINE_OR_MODULE:** Single Canonical Emission Path
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / partial implementation
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers

- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend ELISENCE Phase 4 — Golden Capability & Architecture References
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED Internal Golden Operational References (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-24 — BBVP Milestones M1–M10 Execution Plan

- **PHASE4_CAPABILITY_ID:** P4–G–24
- **CAPABILITY_NAME:** BBVP Milestones M1–M10 Execution Plan
- **CAPABILITY_DOMAIN:** Programme Plan
- **ORIGINAL_PURPOSE:** Foundation→SecureWrapper→FHIR→Privacy→Warehouse→FL→CDS→Portals→Twins→SEAL
- **WHAT_IT_DOES:** Foundation→SecureWrapper→FHIR→Privacy→Warehouse→FL→CDS→Portals→Twins→SEAL
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Programme
- **RELATED_ENGINE_OR_MODULE:** BBVP Milestones M1–M10 Execution Plan
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-G-25 — GHX DONE Definition (100% + Evidence)

- **PHASE4_CAPABILITY_ID:** P4–G–25
- **CAPABILITY_NAME:** GHX DONE Definition (100% + Evidence)
- **CAPABILITY_DOMAIN:** Acceptance Gate
- **ORIGINAL_PURPOSE:** 99% ≠ DONE
- **WHAT_IT_DOES:** 99% ≠ DONE; requires emitters from listed phases, SecureWrapper path, ≥3 CDS demos, ≥2 countries, FL PoC ≥2 nodes, Digital Twin for ≥1 disease, Gov portal 5-lang+RTL, all tests green + SEAL
- **PRIMARY_USERS_OR_CONSUMERS:** See GHX actors/modes and product contracts
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Programme
- **RELATED_ENGINE_OR_MODULE:** GHX DONE Definition
- **INPUTS:** Documented per component
- **OUTPUTS:** Documented per component
- **DEPENDENCIES:** GHX tier ordering
- **PRIVACY_OR_SECURITY_CONTROLS:** Consent/AKAC/DP/mode as applicable
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/ledger as applicable
- **RESILIENCE_CONTROLS:** Validator block / OFF gate

- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO + RTL where applicable
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (completion criteria, not evidenced as met in golden-spec incompleteness)
- **REPORTED_TEST_STATE:** See سندGHX tier statuses / GHX. Report 164/164 claim (conflicts)
- **REPORTED_ACCEPTANCE_STATE:** Subject to GHX DONE definition
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** SEAL pack designed; artefact not in folder
- **SOURCE_DOCUMENTS:** Phase4-GHX.docx; Phase4-GHX.pdf; سندGHX.pdf; GHX-report for Deck.pdf; GHX. Report.pdf
- **SOURCE_SECTIONS:** GHX-report for Deck items; Phase4-GHX.docx §§4-8; سندGHX tiers
- **KNOWN_LIMITATIONS:** Later tiers planned; OCR issues in PDF; 164/164 conflicts with incomplete tiers
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Mostly institutional/research/backend
- **FUTURE_REACTIVATION_CHECK:** CODE_LOCATED → DEPENDENCIES_VALIDATED → ISOLATED_TESTED → MOUNTED → FEATURE_GATED (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH for interoperability, government portal, research export, privacy, multilingual

P4-I-01 — WORM Ledger Pillar (Investor Trust Narrative)

- **PHASE4_CAPABILITY_ID:** P4-I-01
- **CAPABILITY_NAME:** WORM Ledger Pillar (Investor Trust Narrative)
- **CAPABILITY_DOMAIN:** Audit
- **ORIGINAL_PURPOSE:** Immutable auditability for national trust
- **WHAT_IT_DOES:** Immutable auditability for national trust
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** WORM Ledger Pillar (Investor Trust Narrative)
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / claimed validated
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-02 — FHIR Interoperability Pillar

- **PHASE4_CAPABILITY_ID:** P4-I-02
- **CAPABILITY_NAME:** FHIR Interoperability Pillar
- **CAPABILITY_DOMAIN:** Interoperability
- **ORIGINAL_PURPOSE:** FHIR Medication/MedicationRequest/MedicationStatement + standards

- **WHAT_IT_DOES:** FHIR Medication/MedicationRequest/MedicationStatement + standards
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** FHIR Interoperability Pillar
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-03 — Advanced Provenance Engine

- **PHASE4_CAPABILITY_ID:** P4-I-03
- **CAPABILITY_NAME:** Advanced Provenance Engine
- **CAPABILITY_DOMAIN:** Provenance
- **ORIGINAL_PURPOSE:** End-to-end integrity: privacy level/AKAC state/QA results/WORM hash
- **WHAT_IT_DOES:** End-to-end integrity: privacy level/AKAC state/QA results/WORM hash
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** Advanced Provenance Engine
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe

- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-04 — AKAC Adaptive Knowledge Access Control

- **PHASE4_CAPABILITY_ID:** P4-I-04
- **CAPABILITY_NAME:** AKAC Adaptive Knowledge Access Control
- **CAPABILITY_DOMAIN:** Privacy
- **ORIGINAL_PURPOSE:** Adaptive k-anonymity + cross-jurisdiction policy mapping
- **WHAT_IT_DOES:** Adaptive k-anonymity + cross-jurisdiction policy mapping
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** AKAC Adaptive Knowledge Access Control
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / claimed heavily tested
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (default future states start at NOT_YET_REDISCOVERED unless code mission updates)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-05 — Drug Effectiveness Intelligence

- **PHASE4_CAPABILITY_ID:** P4-I-05
- **CAPABILITY_NAME:** Drug Effectiveness Intelligence
- **CAPABILITY_DOMAIN:** Effectiveness Analytics
- **ORIGINAL_PURPOSE:** Mean effectiveness curves, response rates, NBI inputs
- **WHAT_IT_DOES:** Mean effectiveness curves, response rates, NBI inputs
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** Drug Effectiveness Intelligence
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots

- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-06 — Safety & Adverse-Event Intelligence (PRR/ROR)

- **PHASE4_CAPABILITY_ID:** P4-I-06
- **CAPABILITY_NAME:** Safety & Adverse-Event Intelligence (PRR/ROR)
- **CAPABILITY_DOMAIN:** Safety Analytics
- **ORIGINAL_PURPOSE:** MedDRA-coded events; PRR/ROR/severity/time/geo themes
- **WHAT_IT_DOES:** MedDRA-coded events; PRR/ROR/severity/time/geo themes
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** Safety & Adverse-Event Intelligence (PRR/ROR)
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-07 — Comparative Intelligence Engine / NBI

- **PHASE4_CAPABILITY_ID:** P4-I-07
- **CAPABILITY_NAME:** Comparative Intelligence Engine / NBI
- **CAPABILITY_DOMAIN:** Comparative Analytics
- **ORIGINAL_PURPOSE:** Drug-vs-drug comparisons; Net Benefit Index; propensity; multilingual narratives

- **WHAT_IT_DOES:** Drug-vs-drug comparisons; Net Benefit Index; propensity; multilingual narratives
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** Comparative Intelligence Engine / NBI
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (flagship marketing module)
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-08 — National-Scale Population Intelligence

- **PHASE4_CAPABILITY_ID:** P4-I-08
- **CAPABILITY_NAME:** National-Scale Population Intelligence
- **CAPABILITY_DOMAIN:** Population Health
- **ORIGINAL_PURPOSE:** National drug safety surveillance / formulary decision support framing
- **WHAT_IT_DOES:** National drug safety surveillance / formulary decision support framing
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** National-Scale Population Intelligence
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe

- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface ELISENCE Phase 4 — Golden Capability & Architecture Reference
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount Internal Golden Operational Reference (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-09 — QA Gates v2

- **PHASE4_CAPABILITY_ID:** P4-I-09
- **CAPABILITY_NAME:** QA Gates v2
- **CAPABILITY_DOMAIN:** Quality
- **ORIGINAL_PURPOSE:** Completeness, recency, duplicates, outliers, consistency, privacy thresholds before release
- **WHAT_IT_DOES:** Completeness, recency, duplicates, outliers, consistency, privacy thresholds before release
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** QA Gates v2
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-10 — Anti-Bias Mechanism Pack (8 controls)

- **PHASE4_CAPABILITY_ID:** P4-I-10
- **CAPABILITY_NAME:** Anti-Bias Mechanism Pack (8 controls)
- **CAPABILITY_DOMAIN:** Scientific Integrity
- **ORIGINAL_PURPOSE:** k≥50, stratified segmentation, demographic normalization, geographic balancing, outlier removal, propensity correction, bootstrap CIs, AKAC constraints
- **WHAT_IT_DOES:** k≥50, stratified segmentation, demographic normalization, geographic balancing, outlier removal, propensity correction, bootstrap CIs, AKAC constraints
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** Anti-Bias Mechanism Pack (8 controls)
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance

- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-11 — Precomputed Aggregate Low-Latency API Path

- **PHASE4_CAPABILITY_ID:** P4-I-11
- **CAPABILITY_NAME:** Precomputed Aggregate Low-Latency API Path
- **CAPABILITY_DOMAIN:** Performance Architecture
- **ORIGINAL_PURPOSE:** Precomputed aggregates; claimed <200ms / 20–120ms load-tested at 10M+ users
- **WHAT_IT_DOES:** Precomputed aggregates; claimed <200ms / 20–120ms load-tested at 10M+ users
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** Precomputed Aggregate Low-Latency API Path
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED / performance claimed
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-I-12 — Archive & Snapshot Engine with WORM Signatures

- **PHASE4_CAPABILITY_ID:** P4-I-12
- **CAPABILITY_NAME:** Archive & Snapshot Engine with WORM Signatures
- **CAPABILITY_DOMAIN:** Resilience / Audit

- **ORIGINAL_PURPOSE:** Archive/snapshot with WORM signatures; horizontal scaling posture
- **WHAT_IT_DOES:** Archive/snapshot with WORM signatures; horizontal scaling posture
- **PRIMARY_USERS_OR_CONSUMERS:** Governments, clinicians, regulators, citizens (as claimed consumers)
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Investor Trust / Comparative Intelligence narrative
- **RELATED_ENGINE_OR_MODULE:** Archive & Snapshot Engine with WORM Signatures
- **INPUTS:** Analytics inputs
- **OUTPUTS:** Intelligence outputs / APIs
- **DEPENDENCIES:** Often overlaps Analytics/GHX but branding ALISEN
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM/provenance
- **RESILIENCE_CONTROLS:** Snapshots
- **LANGUAGE_SUPPORT:** EN/FA/AR/TR/RO narratives claimed
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED
- **REPORTED_TEST_STATE:** Qualitative “heavily tested/load-tested”; no suite dump
- **REPORTED_ACCEPTANCE_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_RUNTIME_STATE:** DOCUMENTARY STATUS UNCLEAR
- **REPORTED_SEAL_STATE:** DOCUMENTARY STATUS UNCLEAR
- **SOURCE_DOCUMENTS:** Phase4- investor report.pdf; Phas4-investor issue report.pdf; Phase4-investor slide.pdf; Phase4-executive.pdf
- **SOURCE_SECTIONS:** Investor report §§2-6; issue report §§2-5
- **KNOWN_LIMITATIONS:** May over-claim vs Analytics 12-module set; absolute compliance language unsafe
- **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Intended national intelligence surface
- **FUTURE_REACTIVATION_CHECK:** NOT_YET_REDISCOVERED → reconcile with P4-A/P4-G before mount (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
- **ALBANIA_EVIDENCE_RELEVANCE:** HIGH interest but claim discipline required

P4-M-01 — Phase4×Phase8 Freeze/Scan/Map/Rebuild/Integrate Procedure

- **PHASE4_CAPABILITY_ID:** P4-M-01
- **CAPABILITY_NAME:** Phase4×Phase8 Freeze/Scan/Map/Rebuild/Integrate Procedure
- **CAPABILITY_DOMAIN:** Future Reactivation Process
- **ORIGINAL_PURPOSE:** Binding merge method
- **WHAT_IT_DOES:** Stage0 freeze+SHA256; Stage1 scan APIs/schemas/KPIs/UI/RBAC/tests/DB; Stage2 REUSE/REFACTOR/REBUILD/DROP; Stage3 rebuild with governance; Stage4 integration gate; Stage5 operational declaration
- **PRIMARY_USERS_OR_CONSUMERS:** Integration team
- **ARCHITECTURAL_OWNER_OR_SUBSYSTEM:** Post Phase 8 Roadmap
- **RELATED_ENGINE_OR_MODULE:** Merge procedure
- **INPUTS:** Phase4+Phase8 artefacts
- **OUTPUTS:** Mapping tables, evidence, declaration
- **DEPENDENCIES:** Phase 8 cores/engines DONE prerequisites
- **PRIVACY_OR_SECURITY_CONTROLS:** AKAC in gate
- **AUDIT_OR_PROVENANCE_CONTROLS:** WORM in gate
- **RESILIENCE_CONTROLS:** N/A
- **LANGUAGE_SUPPORT:** N/A
- **REPORTED_IMPLEMENTATION_STATE:** DOCUMENTED AS DESIGNED (checklist unchecked)
- **REPORTED_TEST_STATE:** Requires pytest/OFF/ON/RBAC on rebuild
- **REPORTED_ACCEPTANCE_STATE:** Integration gate checkboxes
- **REPORTED_RUNTIME_STATE:** NOT CURRENT
- **REPORTED_SEAL_STATE:** N/A
- **SOURCE_DOCUMENTS:** Plan after cores and engines. Phase 8.pdf/.docx
- **SOURCE_SECTIONS:** Stages 0–5

- ELISENCE Phase 4 — Golden Capability & Architecture Reference
- Internal Golden Operational Reference
- **KNOWN_LIMITATIONS:** Defines Phase 4 as Observability & Product Shell — conflicts with GHX backbone definition
 - **CURRENT_VISIBILITY_IN_MODERN_ELISENCE_IF_DOCUMENTED:** Process
 - **FUTURE_REACTIVATION_CHECK:** Process to execute when reactivating (*default future states start at NOT_YET_REDISCOVERED unless code mission updates*)
 - **ALBANIA_EVIDENCE_RELEVANCE:** MEDIUM — integration hygiene

7. ANALYTICS & HEALTH INTELLIGENCE

Phase 4 analytics are explicitly **population-level, descriptive, non-diagnostic**. Core families: utilization, effectiveness (conditional), satisfaction, adherence, AE radar, plus investor comparative/NBI narrative. Global Insights connectors make intelligence consumable by institutional dashboards.

8. POPULATION HEALTH & EFFECTIVENESS

Utilization answers “how is therapy use evolving?” Effectiveness answers “what Δ weight profiles are observed across anonymised cohorts?” — explicitly **not** “is this drug right for this patient?” (Phase4-Report.pdf §2.3). National-scale population intelligence appears strongly in Identity C documents.

9. MEDICATION / TERMINOLOGY / INTEROPERABILITY

- RxNorm, ATC/DDD, MedDRA, GLP-1 alias mapping (Analytics)
- FHIR R5 resources + SNOMED/LOINC/RxNorm/ATC (GHX)
- OMOP CDM research compatibility (Analytics + GHX research contracts)
- UK Core / US Core / International FHIR validation (GHX)

10. RESEARCH & EVIDENCE

- Research RBAC role + API keys (Analytics)
- OMOP / FHIR NDJSON / cohort/evidence packs (GHX Product C)
- Machine-checkable AC + pytest auditor replay intention (Analytics)
- Multilingual PDF evidence packs for grants/boards/regulators (Analytics)
- Synthetic data for reproducible QA (Analytics)

11. PUBLIC / GOVERNMENT / INSTITUTIONAL CAPABILITY

Surface	Source identity
Public anonymised dashboards	Analytics RBAC Public
University research access	Analytics Research; GHX Research feeds
Ministry portals / heatmaps / foresight	GHX Product A; Investor ministry sections
Hospital BI / ward / EHR CDS	Analytics connectors; GHX Products B + CDS
Admin WORM/config/export	Analytics Admin

12. PRIVACY ARCHITECTURE

- K≥50 public cohorts; insufficient_data below threshold
- Differential privacy flags/noise
- AKAC adaptive k + DP manager (GHX/Investor)
- Ministry/research exports aggregated/de-identified
- Consent-gated export path (GHX)

13. ZERO TRUST / RBAC / AUTHORITY

ELISENCE Phase 4 — Golden Capability & Architecture Reference

Internal Golden Operational Reference

- Zero-Trust stated as foundation (Analytics + GHX)
- Analytics roles: Public / Research / Admin
- GHX actors/modes with explicit denial error codes
- Purpose-bound consent verification before export
- Feature-flag hard OFF (GHX)

14. AUDIT / PROVENANCE / WORM

- Analytics WORM: exports, KPI publications, config changes, incidents
- GHX hash-chained ledger for exports/policy decisions
- Investor Advanced Provenance (QA/AKAC/WORM hash metadata)

15. DATA CONTRACTS / SCHEMA / VERSIONING

- `schema_version`, `kpi_version` validation
- Versioned JSON Schemas / data contracts
- Consent schema fields (`consentID`, `purpose`, `jurisdictionRules`, ...)
- Versioned policy registry packs

16. CONNECTORS / GLOBAL INSIGHTS

- Global Insights read-only APIs (≥ 3 charts, ≥ 3 KPI cards)
- Hospital BI / ministry / academic portal consumption
- GHXQueue routing to warehouse/CDS/export
- ISecureDataEmitter as mandatory cross-phase connector

17. RESILIENCE / FAILURE SAFETY

- Circuit Breaker
- Serve-Last-Good
- Read-Only mode
- Backfill/recovery
- FHIRValidator block-on-invalid
- OFF denial without fake output
- Guarantees claimed: no corruption, no PII leak, continue safe snapshots

18. OBSERVABILITY / MONITORING

- Latency / error / throughput metrics
- Prometheus/Grafana mentioned as example stack
- Incident events into WORM

19. SYNTHETIC DATA / TEST INFRASTRUCTURE

- Synthetic generators for QA/demos
- pytest auditor replay design
- Failure-mode simulations
- Schema validation tests
- GHX BBVP “VERIFY: pytest -q MUST be 100% green”

20. MULTILINGUAL / RTL

ELISENCE Phase 4 — Golden Capability & Architecture Reference

Internal Golden Operational Reference

Language	Documented
English	Yes
Persian (FA)	Yes + RTL
Arabic (AR)	Yes + RTL; Arabic-first localisation noted in GHX
Turkish (TR)	Yes
Romanian (RO)	Yes

Surfaces: PDF export, CDS cards, portals, i18n registry, cultural baselines.

21. REPORTING / EXPORT

- Multi-language PDF export engine
- Research evidence packs (schema/codebooks/consent/WORM proofs)
- SEAL evidence ZIP structure (GHX)
- API KPI/chart payloads

22. REGULATORY-AWARE / MEDICAL-AUTHORITY BOUNDARIES

Layer	Meaning
Technical capability	Documented engines/controls
Clinical authority	Not claimed — non-diagnostic posture
Regulatory approval	Not evidenced — alignment language only (NHS/FDA/MDR/GDPR/PDPL/HIPAA)

23. COMPLETE TEST HISTORY

23.1 Analytics suite (~244)

Source: Phase4–Report.pdf §3 title “Testing & Evidence — 244 Passing Tests”; also status header and §7 (~244).

Categories contributing: - Unit tests - Integration tests - Failure mode simulations - Privacy & k-anon tests - AC-1..AC-20 acceptance tests - Synthetic data & schema validation - Admin / WORM / RBAC checks

Skips: explicit, documented, classified (feature not yet enabled / environment-dependent).

23.2 GHX 164/164 claim

Source: GHX. Report.pdf header “(Status: 164/164 Tests Passed — Fully Verified)”. Scope: non-technical executive claim for GHX.

Conflicts with سندGHX.pdf incomplete Tier 2–4 statuses and Deck “when sealed” conditionality.

23.3 GHX golden-spec tier tests

Source: سندGHX.pdf - Tier 0: implemented + 100% green - Tier 1: skeleton/stub/validator tests green - Tier 2: structures/tests present; full CDS server/UI later - Tier 3–4: planned M4–M9 - Actors/modes: fully covered by tests - Cultural models: implemented with tests

23.4 Investor qualitative tests

“Heavily tested”, “load-tested”, “fully validated” without suite dumps (Phas4–investor issue report.pdf).

Analytics AC themes (AC-1..AC-20)

Isolation; interoperability RxNorm/ATC/MedDRA/OMOP; API latency p95/p99; ETL performance & scalability; k-anon & DP; RBAC & security; WORM & auditability; retention & erasure; observability & monitoring; safe merge/deployment path; etc. (Phase4-Report.pdf §2.11).

Go/No-Go endpoint: overall_status: "G0" | "N0-G0" + per-AC + evidence refs.

GHX DONE definition

100% + Evidence only; listed operational gates including CDS demos, 2 jurisdictions, FL PoC, Digital Twin, Gov portal 5-lang+RTL, SEAL ZIP (Phase4-GHX.docx §8).

25. HISTORICAL IMPLEMENTATION / TEST / ACCEPTANCE STATUS MAP

Identity	Implemented (docs)	Tested (docs)	Accepted (docs)	Sealed (docs)
Analytics Engine	Claimed completed	~244 claimed	Harness exists; GO result not found	Not found
GHX	Partial by tier	164/164 claimed vs tier greens	DONE criteria defined	Pack designed; artefact absent
Investor intelligence	Designed/claimed	Qualitative only	Unclear	Unclear

26. HUMA-LIKE ARCHITECTURAL CAPABILITY MAP

Internal mapping only — **not** a superiority claim vs Huma.

Common health-platform problem space	Phase 4 documentary overlap
Longitudinal / remote health signals	Multi-channel sensor/voice/manual into analytics; twin contracts
Population health	Utilization, national intelligence, ministry portals
Analytics & evidence	Global Analytics Engine; PDF evidence packs
Chronic-condition insights	GLP-1 effectiveness/AE/adherence; diabetes/cardiac twins (contracts)
Institutional health intelligence	Global Insights connectors; hospital/ministry products
Research	OMOP/FHIR NDJSON; Research RBAC; synthetic QA
Healthcare data continuity / interoperability	Vocabulary mapping; FHIR transform/validate; single emission path

27. ALBANIA / EIT RELEVANT EVIDENCE

Theme	Capability IDs	Safe claim	Must not claim
Academic research	P4-A-01,08,12,13,20; P4-G-20	Documented research-oriented privacy-preserving aggregate/export architecture	“Already running Albania study”
Clinical evaluation (non-diagnostic)	P4-A-03,05,06; P4-I-05..07	Documented population effectiveness/adherence/AE analytics design	Clinical superiority / treatment guidance
Population / public health	P4-A-02; P4-I-08; P4-G-18	Documented utilisation & ministry dashboard contracts	Deployed national surveillance
Privacy-preserving analytics	P4-A-07; P4-G-09; P4-I-04	Documented k-anon/DP/AKAC controls	Legal certification

Theme	Capability IDs	Safe claim	Must not claim
ELISENCE Phase 4 — Golden Capability & Architecture Reference		Internal Golden Operational Reference	
Multilingual infrastructure	P4-A-20; P4-G-02,11	Documented EN/FA/AR/TR/RO + RTL	Perfect localisation proven in Albania
Interoperability	P4-A-01; P4-G-06,07,19	Documented RxNorm/ATC/MedDRA/FHIR/OMOP orientation	Certified EHR integration live
Resilience	P4-A-14..17	Documented circuit breaker/SLG/read-only/backfill	Zero-downtime proven nationally
Governance / trust	P4-A-11,18,19; P4-G-08,12,21	Documented WORM/AC/Go-No-Go/consent/seal design	Sealed & accepted by a named regulator

28. SAFE CLAIMS

- Phase 4 was documented as a deep analytics + exchange/governance programme.
- Documents describe 12 Analytics modules and a broad GHX component architecture.
- Documents assert ~244 Analytics tests and describe AC-1..AC-20 + Go/No-Go.
- Documents describe privacy, RBAC/Zero-Trust, WORM, resilience, multilingual/RTL capabilities.
- Documents consistently frame outputs as non-diagnostic / aggregate.
- A future reactivation checklist can and should track every ledger row.

29. CLAIMS THAT MUST NOT BE MADE

- Implemented ⇒ clinically validated / medically approved
- Privacy-aware ⇒ legally certified compliant
- Regulatory-aware ⇒ regulatory approved / MDR Class cleared
- 244 or 164/164 ⇒ independently re-verified without suites
- Designed seal pack ⇒ seal completed
- Documented architecture ⇒ currently mounted in modern ELISENCE
- Comparative/NBI marketing ⇒ proven identical to Analytics AC-tested modules without code reconciliation
- ELISENCE clinically superior to Huma

30. FUTURE PHASE 4 REACTIVATION MASTER CHECKLIST

For **every** PHASE4_CAPABILITY_ID above, maintain a living status among:

NOT_YET_REDISCOVERED | CODE_LOCATED | DEPENDENCIES_VALIDATED | ISOLATED_TESTED | INTEGRATION_TESTED | MOUNTED | FEATURE_GATED | ENABLED | USER_VISIBLE | INSTITUTION_VISIBLE | RESEARCH_VISIBLE | PRODUCTION_VALIDATED | NOT_APPLICABLE

Initial default for modern project (documentary mission): NOT_YET_REDISCOVERED for all product capabilities, except process row P4-M-01 which is a procedure to execute.

Reactivation is incomplete until: 1. Identity A/B/C reconciliation decision is recorded 2. Every HIGH Albania/EIT-relevant capability is at least CODE_LOCATED or explicitly NOT_APPLICABLE with rationale 3. Privacy/RBAC/WORM/resilience controls are INTEGRATION_TESTED before ENABLE 4. Fresh Go/No-Go or equivalent evidence pack is produced

31. DEPENDENCY / ACTIVATION ORDER MAP

Analytics suggested order

- Isolation boundary + Zero-Trust/RBAC + Privacy Guard
- Vocabulary mapping
- Quality gates / schemas / synthetic data
- Aggregation engines (utilization → adherence/satisfaction/AE → effectiveness if weights ready)

GHX suggested order (aligns M1–M10)

Foundation/Identity/Locale/Policy → Emitter/SecureWrapper/Consent → FHIR transform/validate → Privacy AKAC/DP → Warehouse/Federated query → FL PoC → CDS/Hospital APIs → Gov/Research portals → Digital Twin → SEAL pack

Critical dependency edges

- Effectiveness → Vocabulary + weight data + Privacy Guard
- AE Radar → MedDRA mapping + Privacy Guard
- Global Insights → all KPI engines + RBAC + Privacy
- PDF export → KPI engines + Privacy + WORM
- Research OMOP export → Consent + AKAC/DP + WORM + schemas
- CDS cards → Secure path + Locale/i18n + non-diagnostic rules
- Ministry portal tiles → Privacy + WORM + cultural baselines

32. GOLDEN MASTER INDEX

- Systems/identities:** 3 (Analytics Engine, GHX, Investor Trust/Intelligence) + merge procedure
- Analytics modules:** 12
- GHX deck components:** 20
- Capability ledger rows:** 62
- Test suite families documented:** Analytics (~244 categories), GHX executive 164/164, GHX tier tests, investor qualitative, failure/privacy/schema/RBAC/WORM/language themes
- Acceptance constructs:** AC-1..AC-20; Go/No-Go endpoint; GHX DONE definition; SEAL pack
- Privacy/security controls:** Privacy Guard; AKAC; DP; RBAC; Actors/Modes; ConsentEngine; Zero-Trust; Feature-flag OFF; FHIRValidator block
- Resilience controls:** Circuit Breaker; SLG; Read-Only; Backfill/Recovery; hard OFF denial
- Languages:** EN, FA, AR, TR, RO (+RTL FA/AR)
- Research capabilities:** Research RBAC; OMOP/FHIR NDJSON; evidence packs; synthetic QA; FL contracts
- Institutional capabilities:** Global Insights; ministry portal; hospital/EHR CDS; admin WORM/export
- Outputs/exports:** KPI APIs; charts; PDFs; OMOP/FHIR packs; SEAL ZIP structure
- External connectors:** ISecureDataEmitter; BI portals; EHR FHIR/CDS; federated nodes

33. SOURCE TRACEABILITY INDEX

Source file	Role in golden reconstruction
Phase4-Report.pdf	Primary Analytics architecture/test/AC
Elisence — Phase 4 (Global Analytics Engine).pdf	12-module snapshot
Phase4-خلاصه مدیریتی.pdf	Persian managerial confirmation of modules/AC/Seal language
Phase4-GHX.docx / .pdf	GHX architecture, M1–M10, DONE
سندGHX.pdf	Golden Spec v1.1 tier implementation/test statuses
GHX-report for Deck.pdf	20-component map + SEAL pack definition
GHX. Report.pdf	164/164 executive claim
Phase4- investor report.pdf	Identity C master script

Source file	Role in golden reconstruction
ELISENCE Phase 4 — Golden Capability & Architecture Reference	Internal Golden Operational Reference
Phas4-investor issue report.pdf	NBI/anti-bias/performance claims
Phase4-investor slide.pdf	Compressed trust moats
Phase4- executive.pdf	Multi-phase investor brief including Phase 4 shield framing
Plan after cores and engines. Phase 8.*	Future merge/reactivation procedure
Phase4 Excutive.pdf	EXCLUDED (Phase 6 misfile)

Capabilities with source traceability: **62/62**

34. DOCUMENTARY GAPS / UNCERTAINTIES

1. Codebase not in documentary folder — implementation not independently code-verified
2. No raw pytest logs or CI artefacts for 244 or 164 counts
3. No AC pass/fail table or stored GO certificate
4. No SEAL ZIP present in PHASE4 folder
5. Analytics vs GHX vs Investor identity reconciliation unresolved
6. 244 vs 164 totals unresolved
7. Phase4-GHX.pdf OCR degradation (DOCX preferred)
8. سندGHX extract truncated mid §6.1
9. Exact schema catalogues / endpoint lists not fully enumerated
10. Effectiveness weight-data dependency state unknown historically
11. NBI/comparative engine relationship to Analytics modules unclear
12. Branding Elisence vs ALISEN inconsistency
13. Phase 8 plan defines Phase 4 differently (Observability shell) than GHX backbone

Unresolved documentary gaps counted: 13

35. FINAL GOLDEN PHASE 4 SYSTEM DEFINITION

ELISENCE Phase 4 is the historical programme layer that combines: 1. a **Global Analytics & Evidence Engine** (12 modules) for privacy-preserving population KPIs, institutional dashboards, multilingual evidence export, WORM audit, resilience, and machine-checkable acceptance; and 2. a **GHX Global Health Exchange & Intelligence Gateway** architecture for jurisdiction-aware consent, FHIR interoperability, CDS, federated analytics/learning, and ministry/hospital/research products; with an overlapping investor narrative of **trust, provenance, AKAC, and comparative intelligence (NBI)**.

It was created to produce **safe, auditable, multilingual, non-diagnostic population and institutional intelligence** on top of operational care data — not to replace clinical judgement, and not (on documentary evidence alone) as a currently mounted modern-runtime feature set.

This Golden Reference is the checklist against which future rediscovery, validation, mounting, and Albania/EIT evidence packaging must be measured — **omit nothing in the ledger**.

APPENDIX — OMISSION AUDIT

Rescanned all 15 PHASE4 source files. Material Phase 4 capabilities from Analytics modules 1–12, GHX deck components 1–20, GHX tiers/products/DONE/M1–M10, investor pillars/intelligence/QA/anti-bias/performance, and Phase8 merge procedure were entered into the ledger or explicitly indexed. Phase 6 misfile content was identified and excluded from Phase 4 capabilities. Persian خلاصه module set cross-checked against English snapshot. Investor NBI/comparative capabilities retained as distinct rows (not dropped due to absence from Analytics 12-module list).

OMISSION_AUDIT_PASS = YES